iEnergy

ARTICLE

https://doi.org/10.23919/IEN.2023.0005

Interaction analysis and enhanced design of grid-forming control with hybrid synchronization and virtual admittance loops

Hong Gong[1] and Xiongfei Wang[2] ✉

ABSTRACT

The hybrid power- and voltage-based synchronization control method has shown potential for enhancing the stability of grid-forming (GFM) inverters. However, its effectiveness may be compromised if other control loops are not properly designed. To address the control-loop interactions, this paper presents a design-oriented analysis method for multiloop-controlled GFM inverters. The method begins by identifying the dominant oscillation modes through modal analysis. The sensitivities of damping ratios to control parameters are then determined for the dominant modes, which allows for characterization of control-loop interactions. A co-design method of GFM control is next developed based on the sensitivity analysis. Lastly, simulations and experimental results are presented to confirm the effectiveness of the method.

KEYWORDS

Grid-forming inverter, hybrid synchronization controller, virtual admittance, damping ratio, controller design.

The ever-increasing deployment of inverter-based resources has led to the rise of grid-forming (GFM) technology as a means of improving the stability and resilience of power systems[1–3]. The dynamics analysis of GFM inverters have proven their higher stability robustness than grid-following (GFL) inverters in weak alternating current (AC) grids[4]. However, GFM inverters are prone to instability in strong grids[5]. To address this issue, a hybrid synchronization control (HSC) method that combines voltage- and power-based synchronization loops is recently introduced for GFM inverters[6–8]. The HSC allows for a flexible GFL and GFM configuration, providing a universal control structure for inverters[6]. In addition, a virtual-admittance-based current control is developed to actively shape the impedance of GFM inverter and limit overcurrent, providing an additional degree of control that can help to dampen system oscillations[9–11].

There has been a growing interest in analyzing the effect of control parameters on the stability of GFM inverters. In Ref. [12], the relationship between droop control gains and stability margins is studied, and the impact of droop gains on the stability of GFM inverters is analyzed. The effect of the inner voltage control loop is discussed in Ref. [13], which shows that an increase in the voltage controller gain can lead to harmonic instability issues in stronger grids due to the increase in the frequency region of negative resistance caused by time delay. The impact of the inner current controller on the stability of GFM inverters is analyzed in Ref. [14], which reveals that a higher integral gain of the current controller can decrease the damping ratio of the system. The influences of the virtual admittance and the power-based synchronization on the system stability are discussed in Refs. [15, 16], respectively. However, those studies in Refs. [12–16] are limited to analyzing the influence of individual control parameters and overlook the dynamic interactions of different controllers in the inner and outer loops.

In Ref. [17], the interaction between the inner voltage control loop and the outer power control loop of GFM inverters is discussed. The study reports that the sub-synchronous control interactions may occur in stiff ac grids and are closely linked to the design of the inner voltage control loop. Another work[18] compares three different inner-loop controllers in terms of their interactions with the outer power control loop, and it reports that the inner vector-voltage control can cause low-frequency oscillations, whereas the inner open-loop voltage control does not have these issues. In contrast, Ref. [19] shows that the inner and outer control loops can be decoupled through proper tuning of the inner-loop controller parameters.

However, these studies[17–19] only examine the impact of inner-loop controllers on the low-frequency dynamics of outer loops, ignoring the fact that tuning one sensitive controller parameter can affect the impact of other parameters on system stability, which is particularly important in the case of multiloop control schemes. Additionally, one specific controller parameter may have the opposing effects on the damping ratios of oscillation modes, making it challenging to find a good compromise between them. This highlights the need of further research on the control-loop interaction analysis and the parameter tuning for multiloop-controlled GFM inverters.

This paper, thus, attempts to fill the void by proposing a design-oriented analysis approach for the GFM inverters using HSC and virtual-admittance-based current control. Dynamic interactions of HSC, ac-bus voltage magnitude control (AVC), virtual admittance, and current control loops are considered. First, a single-input single-output (SISO) transfer-function model is derived using the method described in Ref. [20]. Then, the modal analysis is performed on the SISO model to identify the dominant oscillation modes. The sensitivities of damping ratios to controller parameters are then revealed for the dominant modes, allowing for the identification of critical control loops. The damping ratios are further used to reveal how one control loop impacts the stability of the system when influenced by other control loops. A co-design

[1]Vestas, Aarhus 8200, Denmark; [2]KTH Royal Institute of Technology, Stockholm 11428, Sweden
Address correspondence to Xiongfei Wang, xiongfei@kth.se

method of multiloop controller parameters is developed to enhance system stability robustness. Finally, simulations and experiment results are presented to validate the accuracy of the theoretical analysis and design method.

1 System description

1.1 Overall structure

Figure 1 shows a simplified single-line diagram of a three-phase GFM inverter employing the HSC and the virtual-admittance-based current control. The filter inductor is denoted by L_f and the grid impedance is represented by the inductor L_g. A low short-circuit ratio (SCR) of 1.2 is chosen to emulate a weak ac grid, which leads to a grid inductance (L_g) of 23 mH. A filter capacitor is used to filter switching frequency harmonics of the inverter. It is found in Ref. [21] the filter capacitor mainly impacts the system dynamics in the high-frequency range (above 1000 Hz)[21], and hence it is not considered in the analysis of low-frequency dynamics (below 500 Hz) in this study. I_{abc} and U_{sabc} denote the three-phase inductor current vector and voltage vector at the point of coupling (PoC), respectively, while U_{iabc} and U_{gabc} represent the inverter output voltage vector and grid voltage vector, respectively. The dc-link voltage of GFM inverter is assumed constant in this study, which can be maintained by either a front-end converter[22] or an energy storage element[23]. The parameters of the main circuit and of the control system used in this study are listed in Table 1.

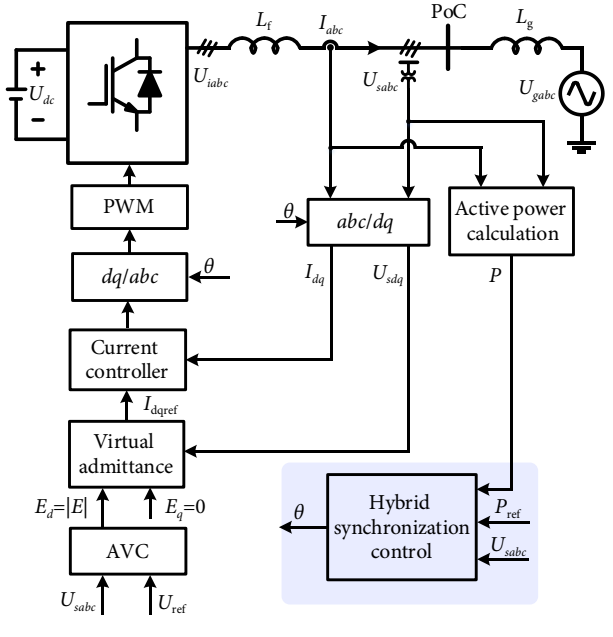


Fig. 1 Simplified single-line diagram of a three-phase grid-forming inverter with the hybrid synchronization control and the virtual-admittance-based current control.

1.2 HSC and AVC

Figure 2 shows the block diagram of HSC. The HSC consists of a voltage-based synchronization (VBS) loop and a power-based synchronization (PBS) loop. The HSC differs from the scheme that adds a conventional phase-locked loop (PLL) at the output of PBS loop[15]. In the HSC, the phase angle fed back to the VBS loop is the sum of outputs of VBS and PBS loops. In the VBS loop, only a proportional gain (K_{pv}) is used, which yields a non-zero phase angle difference between the inverter voltage and PoC voltage.

Table 1 Parameters of the three-phase inverter

Symbol	Description	Value
ω_1	Grid frequency	50 Hz
f_s	Sampling frequency	10 kHz
U_g	Grid voltage (line-to-line)	110 V
I	Phase current (RMS value)	10.5 A
f_{sw}	Switching frequency	10 kHz
V_{dc0}	DC-link voltage	730 V
L_f	Filtered inductor	3 mH
L_g	Grid inductor	23 mH
K_{pc}/K_{ic}	Current inner controller	15.7 / 5480
K_{pv}	Proportional gain of voltage-based synchronization loop	5.71
L_v	Virtual inductance	7.7 mH
R_v	Virtual resistance	1.21 Ω
K_{ppsc}/K_{ipsc}	PI gains of power-based synchronization loop	1.25×10^{-3}/ 7.85×10^{-3}
K_{pac}/K_{iac}	AC voltage magnitude controller	0/100

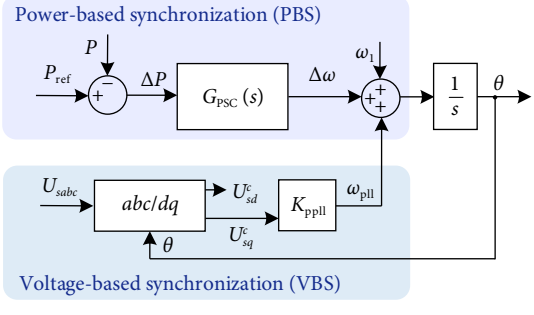


Fig. 2 Block diagram of hybrid synchronization control.

The PBS loop employs a proportional and integral (PI) controller.

The AVC regulates the magnitude of the PoC voltage, $|U_{sabc}|$ to the reference voltage, U_{ref}, generating the d-axis voltage reference, E_d, and setting the q-axis voltage reference, E_q, to zero, to provide the reference input for the virtual admittance controller. While the AVC can be based on a PI controller, only the integrator is adopted in the practical applications.

1.3 Virtual-admittance-based current control

Figure 3 shows the block diagram of the virtual admittance control, which is implemented in the dq frame. The output of the virtual admittance control provides the reference for the vector-current control. The virtual admittance comprises a virtual resistance and inductance in series, which is emulated by using an low-pass filter (LPF) to replace the voltage controller in the traditional dual-loop voltage-vector control scheme[10,9]. Although the virtual admittance in Figure 3 is implemented in the dq-frame, it can also be used directly in the stationary $\alpha\beta$-frame A PI controller is used with the current controller in Figure 1, as it is in the dq-frame. Alternatively, a proportional + resonant (PR) controller is needed in the $\alpha\beta$-frame for a zero steady-state error.

2 Small-signal modelling and validation

2.1 Closed-loop small-signal model

Figure 4 shows the block diagram of the small-signal model of the

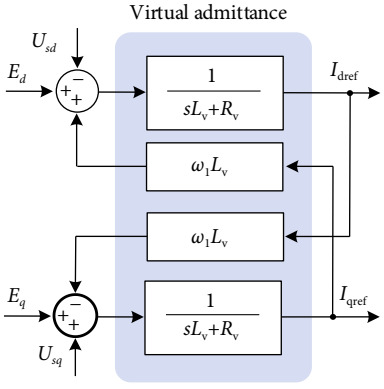


Fig. 3 Block diagram of virtual admittance control in the *dq*-frame.

GFM inverter with the HSC and the virtual-admittance-based current control. The detailed modelling procedure is given in the appendix. Based on Figure 4 and the equations in the appendix, the closed-loop transfer function matrix can be derived, as given by Eq. (1). $\boldsymbol{R}_{ef} = [P_{ref}, U_{ref}]^T$ is the reference of the outer loop, and $\boldsymbol{I}_m$ means the unitary diagonal matrix.

$$[\boldsymbol{G}_d(\boldsymbol{T}^{-1}\boldsymbol{G}_i\boldsymbol{Y}_v\boldsymbol{G}_a - \boldsymbol{T}^{-1}\boldsymbol{G}_i\boldsymbol{Y}_v\boldsymbol{T} - \boldsymbol{T}^{-1}\boldsymbol{G}_i\boldsymbol{Y}_v\boldsymbol{T}\boldsymbol{H}_{12} + \boldsymbol{G}_i\boldsymbol{H}_{22} + \boldsymbol{T}^{-1}\boldsymbol{H}_{32}) - \boldsymbol{I}_m]\boldsymbol{U}^s_{sdq} = [\boldsymbol{Y}_p^{-1} + \boldsymbol{G}_d(\boldsymbol{T}^{-1}\boldsymbol{G}_i\boldsymbol{Y}_v\boldsymbol{T}\boldsymbol{H}_{11} + \boldsymbol{G}_i - \boldsymbol{G}_i\boldsymbol{H}_{21} - \boldsymbol{T}^{-1}\boldsymbol{H}_{31})]\boldsymbol{I}^s_{dq} + \boldsymbol{Y}_p\boldsymbol{G}_d\boldsymbol{G}_i\boldsymbol{Y}_v\boldsymbol{R}_{ef} \quad (1)$$

2.2 Frequency scanning for model validation

To validate the accuracy of the small-signal model, the *dq*-frame admittance model, which maps the relationship between the PoC voltage and inverter current, is derived and compared with the frequency scan result.

Figure 5 shows the frequency-domain responses of Y_{dd}, Y_{dq}, Y_{qd} and Y_{qq} from 1 Hz to 1000 Hz, where the blue lines represent the theoretical admittance model, and the red dots illustrate the measure admittance from a time-domain nonlinear simulation model in MATLAB/Simulink. It is clear that the small-signal model matches well with the measured result.

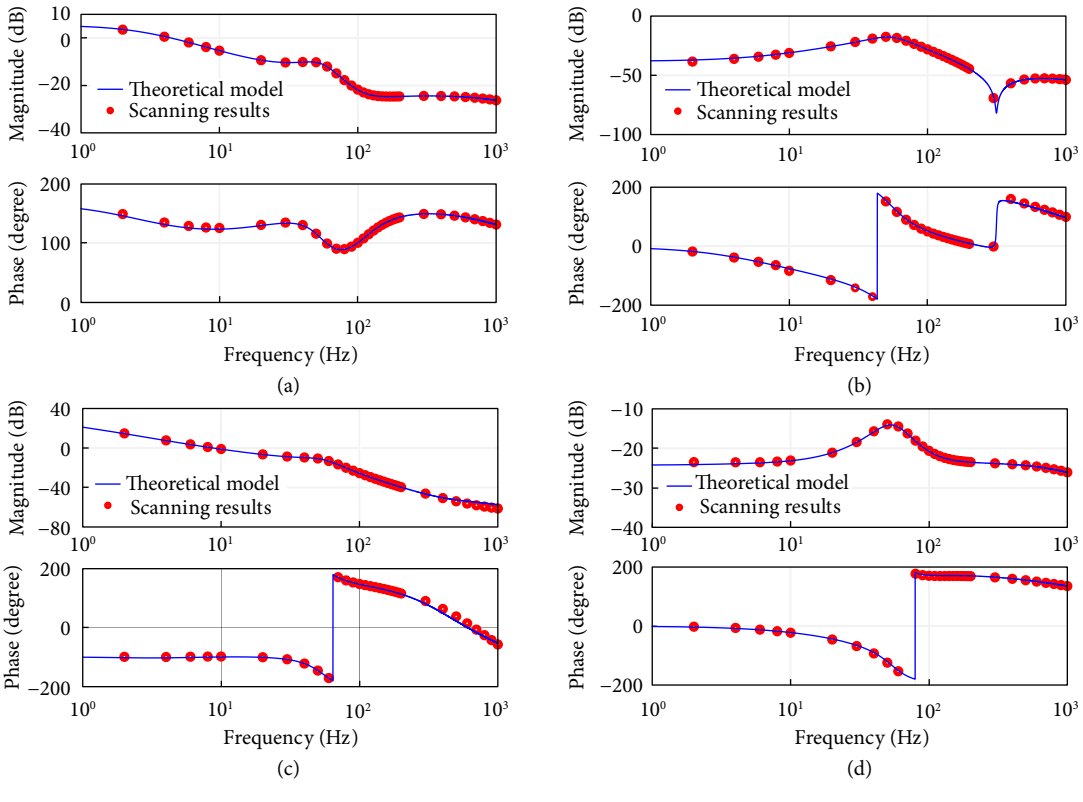


Fig. 5 Validation of theoretical admittance model through frequency scanning. (a) Y_{dd}, (b) Y_{dq}, (c) Y_{qd}, (d) Y_{qq}.

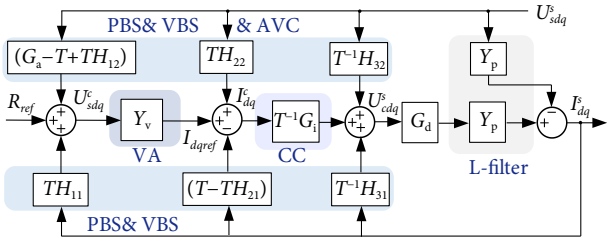


Fig. 4 Block diagram of small-signal model of GFM inverter with the hybrid synchronization control and the virtual-admittance-based current control.

3 Modal analysis of inverter-grid interaction

3.1 Closed-loop model including grid impedance

To analyze the stability of GFM control with varying SCRs of the grid, a small-signal model that incorporates the grid impedance is established[20]. The grid impedance is given as

$$\boldsymbol{Z}_g = \begin{bmatrix} sL_g & -\omega_1 L_g \\ \omega_1 L_g & sL_g \end{bmatrix} \quad (2)$$

where "*s*" is the Laplace variable and ω_1 is the grid fundamental frequency.

The mathematical relationship between the grid voltage and the PoC voltage can be expressed as

$$\boldsymbol{U}_{gdq0}+\boldsymbol{Z}_g\boldsymbol{I}_{dq0}=\boldsymbol{U}_{sdq0} \tag{3}$$

where $\boldsymbol{U}_{sdq}=[U_{sd}, U_{sq}]^T$ is the PoC voltage in the dq-frame and $\boldsymbol{U}_{gdq}=[U_{gd}, U_{gq}]^T$ represents the grid voltage in the dq-frame; $\boldsymbol{I}_{dq}=[I_d, I_q]^T$ denotes the dq-axis current.

Figure 6 shows the block diagram of the small-signal model for the GFM inverter with the grid impedance included. To facilitate the analysis of control-loop interactions, the inner current loop is represented by its closed-loop transfer matrix, $\boldsymbol{G}_c$, which is given as

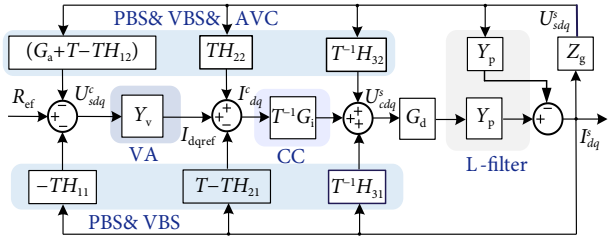


Fig. 6 Block diagram of small-signal model of GFM inverter with the grid impedance.

$$\boldsymbol{G}_c=(\boldsymbol{I}+\boldsymbol{Y}_{pg}\boldsymbol{G}_i\boldsymbol{G}_d)^{-1}\boldsymbol{Y}_{pg}\boldsymbol{G}_i\boldsymbol{G}_d \tag{4}$$

which is a diagonal matrix, since the dq-axis decoupling control is used. The block diagram shown in Figure 6 is also reformulated in Figure 7, where the HSC, the AVC, and the virtual admittance loops are grouped through the block diagram algebra and represented by the interaction matrix $\boldsymbol{G}$.

Figure 8 further simplifies the block diagram shown in Figure 7 with a multiple-input multiple-output (MIMO) representation of the interaction matrix $\boldsymbol{G}$. G_{c1} and G_{c2} are the diagonal entries of transfer matrix $\boldsymbol{G}_c$. G_{11}, G_{12}, G_{21}, G_{22} represent the elements of the interaction matrix. From Figure 8, two SISO closed-loop models[20,24] can be derived as

$$\frac{I'_d}{I'_{dref}}=\frac{G_{c1}G_{11}+G_{c1}G_{c2}\left[G_{11}G_{22}-G_{12}G_{21}\right]}{CE} \tag{5}$$

$$\frac{I'_q}{I'_{qref}}=\frac{G_{c2}G_{22}+G_{c1}G_{c2}\left[G_{11}G_{22}-G_{12}G_{21}\right]}{CE} \tag{6}$$

where the denominators of the two transfer functions, i.e., the characteristic equation (CE) are the same, which is given as

$$\begin{aligned}CE&=1+G_{c1}G_{11}+G_{c2}G_{22}+G_{c1}G_{c2}\left[G_{11}G_{22}-G_{12}G_{21}\right]\\&=1+G_{c1}G_{11}+G_{c2}G_{22}+G_{c1}G_{c2}\det(\boldsymbol{G})\end{aligned} \tag{7}$$

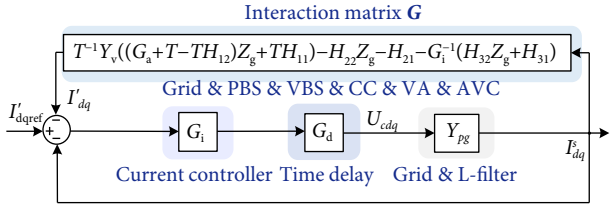


Fig. 7 Equivalent block diagram of GFM inverter with the grid impedance.

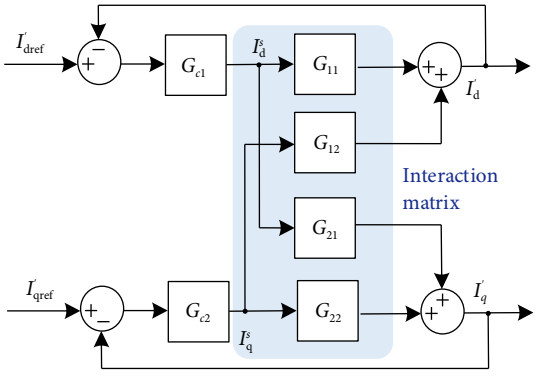


Fig. 8 Equivalent block diagram for control-loop interaction analysis.

where the interaction matrix $\boldsymbol{G}$ is derived as

$$\begin{aligned}\boldsymbol{G}=&\boldsymbol{T}^{-1}\boldsymbol{Y}_v\left((\boldsymbol{G}_a+\boldsymbol{T}-\boldsymbol{TH}_{12})\boldsymbol{Z}_g-\boldsymbol{TH}_{11}\right)\\&-\boldsymbol{H}_{22}\boldsymbol{Z}_g-\boldsymbol{H}_{21}-\boldsymbol{G}_i^{-1}(\boldsymbol{H}_{32}Z_g+\boldsymbol{H}_{31})\end{aligned} \tag{8}$$

3.2 Identification of dominant modes

The zeros of CE indicate the oscillation modes of the system[20,24], which are similar to the eigenvalue analysis based on the state-space model[25].

Figure 9 shows the pole-zero map of the CE of the studied GFM inverter system. Given a complex zero $Z_i=\sigma_i+j\omega_i$ of the CE, its damping ratio ξ_i and oscillatory frequency f_i are given by

$$\xi_i=-\frac{\sigma_i}{\sqrt{\sigma_i^2+\omega_i^2}},\qquad f_i=\frac{\omega_i}{2\pi} \tag{9}$$

As the oscillation modes that are unstable or under-damped dominate the system dynamics when subjected to a disturbance[25], the following modes are considered as dominant modes:

If $\sigma_i>0$ or $\xi_i<0$, the corresponding mode is unstable.

The under-damped mode has a damping ratio $\xi_i<10\%$[26].

From Figure 9, two dominant (under-damped) modes are chosen based on the condition of $\xi_i<10\%$, whose zeros are $-25\pm j330$ and $-2\pm j20$, respectively. Based on Eq. (9), the oscillation frequency

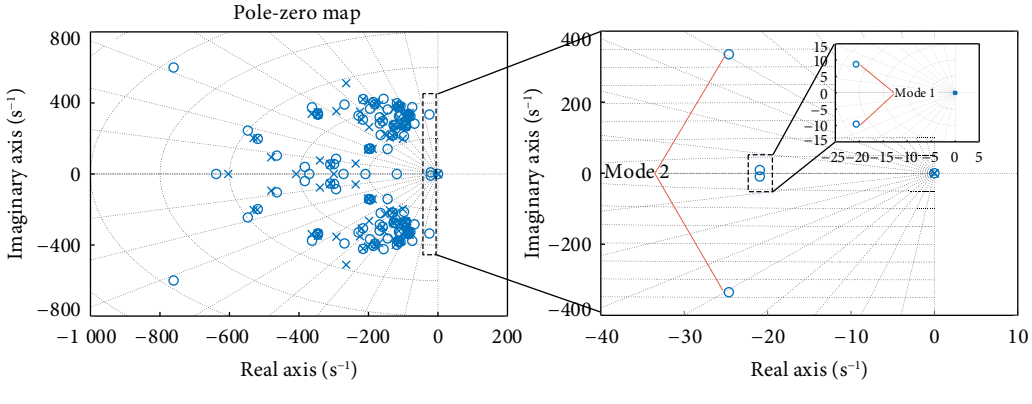


Fig. 9 Pole-zeros map of the characteristic equation of GFM inverter system.

can be calculated from the imaginary part of zeros, which are 3.2 Hz (Mode 1) and 53 Hz (Mode 2), respectively. Thus, the two complex conjugate zeros are used to examine the control-loop interactions, which will be discussed in Section 5.

Figure 10 shows the pole-zero map of the *CE* of the stuidied GFM inverter system with varying SCRs. The damping ratio of Mode 1 decreases as the SCR increases (stronger grid) while the damping ratio of Mode 2 increases. This brings a trade-off for controller parameter selection to ensure positive damping ratios for both modes. Therefore, the lowest and highest SCR of the system should be acounted to ensure worst-case scenarios are considered in the controller parameter tuning.

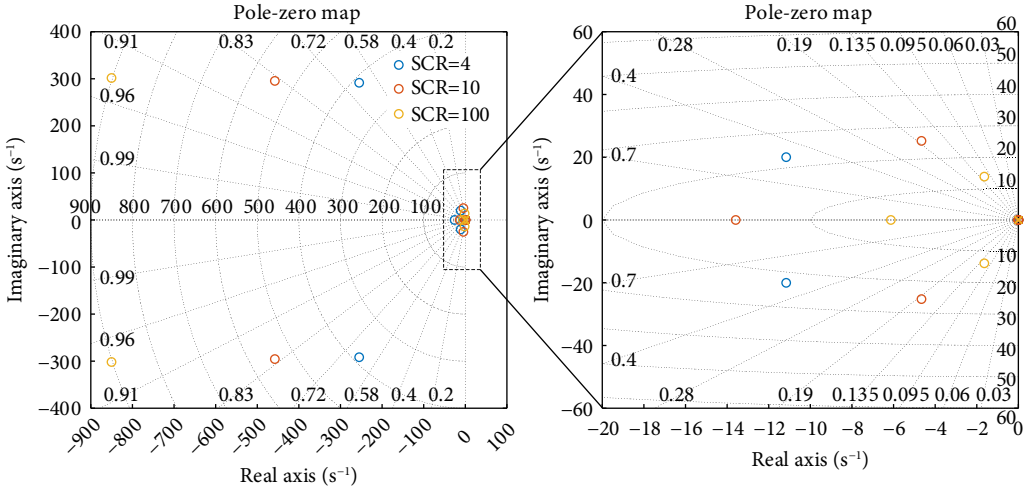


Fig. 10 Pole-zeros map of the characteristic equation of GFM inverter system under different SCRs.

3.3 Sensitivity analysis

To analyze the impact of control loops on the two dominant modes, the parametric sensitivity analysis is performed on the damping ratios of the two modes.

The damping ratio ξ_i with respect to the control parameter K_i can be obtained as[26]

$$\frac{\partial \xi_i}{\partial K_i} \approx \frac{\xi_i(K_{i0}+\Delta K_i)-\xi_i(K_{i0})}{\Delta K_i} \tag{10}$$

where K_i means the i-th parameter (K_{pc}, K_{ic}, L_v, R_v, …). K_{i0} is the initial controller parameter, and $\Delta K_{i0}=10\%K_{i0}$ is the parameter perturbation.

Figure 11 illustrates the damping-ratio sensitivity of Mode 1 (3.2 Hz) to controller parameters. The virtual admittance and the proportional gain of the PBS loop mainly affect the damping ratio of Mode 1. Increasing the virtual resistance R_v decreases the damping ratio of Mode 1, while a larger virtual inductance L_v and a larger proportional gain of the PBS loop increase it.

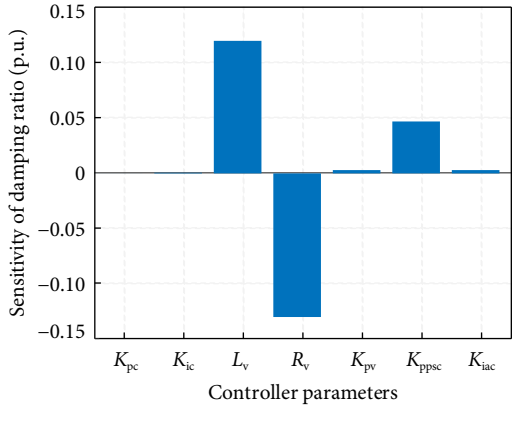


Fig. 11 Sensitivity of damping ratio to controller parameters for the Mode 1.

Figure 12 depicts the damping-ratio sensitivity of Mode 2 (53 Hz) to controller parameters. All controller parameters have a significant effect. Increasing the virtual resistance and the proportional gain of the current controller improves the damping ratio of Mode 2, while a larger virtual inductance, proportional gain of the VBS loop, proportional gain of PBS loop, or integral gain of the AVC decreases it.

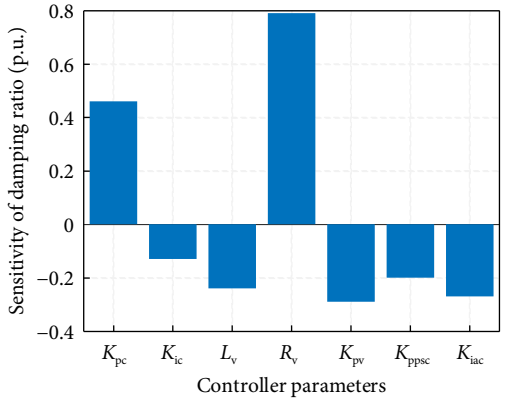


Fig. 12 Sensitivity of damping ratio to controller parameters for the Mode 2.

4 Design-oriented analysis of loop interactions

This section presents a design-oriented analysis of control-loop interactions for the GFM inverter system. The interaction between virtual admittance and PBS can excite the oscillation Mode 1 (3.2 Hz), while interactions between the HSC, AVC, virtual admittance, and the current controller can manifest the oscillation Mode 2 (53 Hz). Further details are discussed below.

4.1 Impact of controller parameters on damping ratios

Figure 13 plots the damping ratio of Mode 2 along with the

change of integral gain of AVC (from K_{iac} to $4K_{iac}$), with other controller parameters kept constant. It is clear that increasing the integral gain of AVC decreases the damping ratio of Mode 2. When the integral gain is above $2.2K_{pac}$, the damping ratio becomes negative, indicating system instability. Since this parameter has little effect on the damping ratio of Mode 1, its change is not presented.

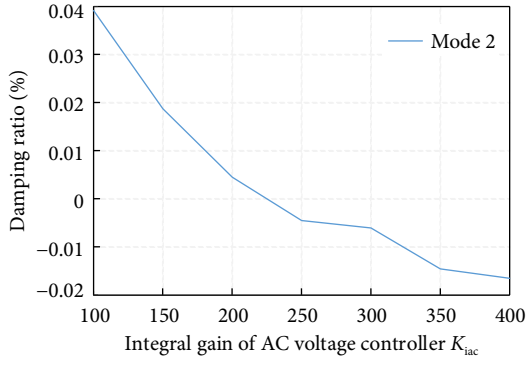


Fig. 13 Effect of K_{iac} on the damping ratio of Mode 2.

Figure 14 shows the effect of changing the proportional gain of the PBS loop (from K_{ppsc} to $4K_{ppsc}$) on the damping ratio of Mode 2. The increase in proportional gain of PBS reduces the damping ratio of Mode 2. If the proportional gain is larger than $2.3K_{ppsc}$, the damping ratio becomes negative, resulting in system instability. The damping ratio change for Mode 1 is not shown, as it remains above 10% with this range of proportional gain of PBS loop.

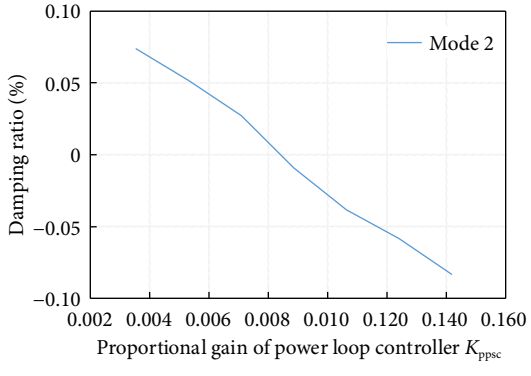


Fig. 14 Effect of K_{ppsc} on the damping ratio of Mode 2.

Figure 15 depicts the damping ratio of the system at Mode 2 with the change of the proportional gain of the current controller (from $0.5K_{pc}$ to $2K_{pc}$). It is shown that the increase of the proportional gain of the current controller would increase the damping ratio of the system at Mode 2. Thus, the impact of the current controller on the system stability should be considered and the assumption that the current controller can be regarded as the unity gain is invalid for the small-signal stability analysis of grid-connected GFM inverter.

Figure 16 shows the effect of changing the proportional gain of the VBS control (from $0.4K_{pv}$ to $2.8K_{pv}$) on the damping ratio of Mode 2. The increase in proportional gain of the VBS control reduces the overall damping ratio of Mode 2. Apparently, the adoption of VBS complicates loop interactions, and hence, its parameters need to be designed deliberately.

Figure 17 shows the effect of changing the virtual resistance (from R_v to 4 R_v) on the damping ratios of Mode 1 and Mode 2. Increasing virtual resistance reduces the damping ratio of Mode 1

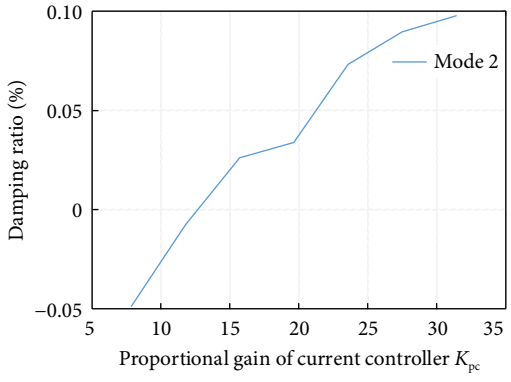


Fig. 15 Effect of K_{pc} on the damping ratio of Mode 2.

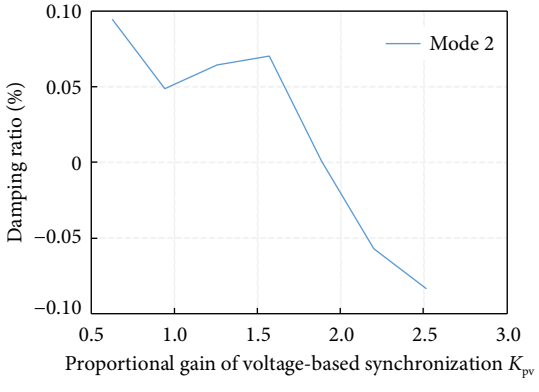


Fig. 16 Effect of K_{pv} on the damping ratio of Mode 2.

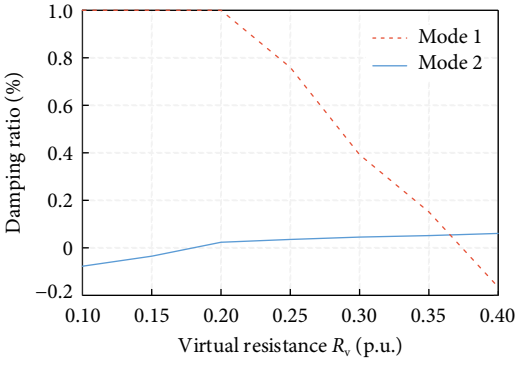


Fig. 17 Effects of R_v on the damping ratios of Mode 1 and Mode 2.

but increases the damping ratio of Mode 2. There is clearly a trade-off in selecting the virtual resistance to ensure the positive damping ratios for both modes.

Figure 18 shows the impact of changing virtual inductance (from $0.3L_v$ to $2L_v$) on the damping ratios of Mode 1 and Mode 2. Unlike virtual resistance, increasing virtual inductance leads to an increase in the damping ratio of Mode 1 but decreases the damping ratio of Mode 2. Thus, there is a trade-off in selecting virtual inductance to secure the positive damping ratios for both modes.

4.2 Control-loop interactions

The control-loop interaction refers to the effect of one control loop on the system stability that is altered by other control loops. This study analyzes the impact of control-loop interactions on system stability by examining changes in the damping ratios of dominant modes.

The effects of control-loop interactions on the damping ratios of dominant modes, specifically Mode 2, are illustrated in Figures

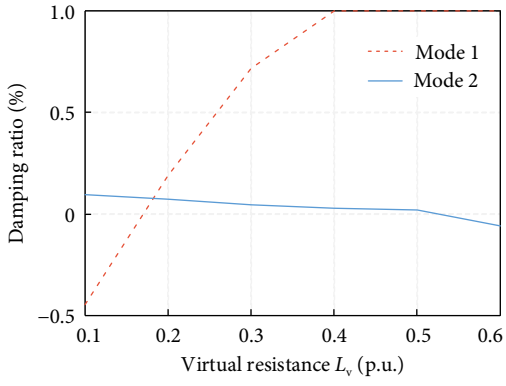


Fig. 18 Effect of L_v on the damping ratios of Mode 1 and Mode 2.

19–21 for different bandwidths of the VBS loop. It is clear that increasing virtual inductance and the bandwidth of VBS loop, while decreasing virtual resistance, leads to a decrease in the damping ratio of Mode 2, as previously predicted by theoretical analysis. Hence, the selection of virtual admittance is a trade-off between improving the damping ratio at Mode 1 (with higher L_v/R_v) and at Mode 2 (with lower L_v/R_v) to maintain system stability.

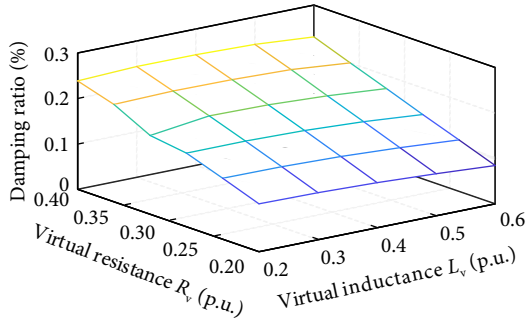


Fig. 19 Effects of changing virtual resistance and virtual inductance on the damping ratio of Mode 2, when the bandwidth of VBS loop is 0 Hz.

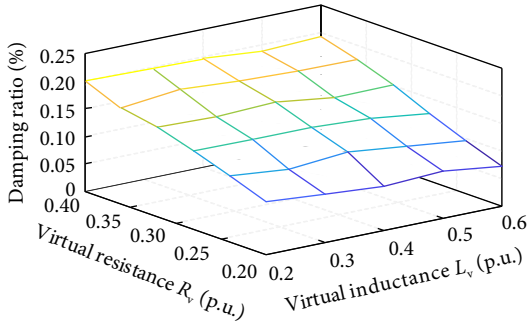


Fig. 20 Effects of changing virtual resistance and virtual inductance on the damping ratio of Mode 2, when the bandwidth of VBS loop is 10 Hz.

4.3 Co-design of multiloop GFM control

Addressing the loop interactions requires a co-design of the multiloop GFM control based on the small-signal modeling. To that end, the GFM inverter system should be initially stable with an equilibirum point for linearization. This requires the initial deisgn of control parameters should make the damping ratios of the system above zero. The dominant oscillation modes are then identified through modal analysis. The critical control loops are next determined based on the damping-ratio sensivitities to the controller

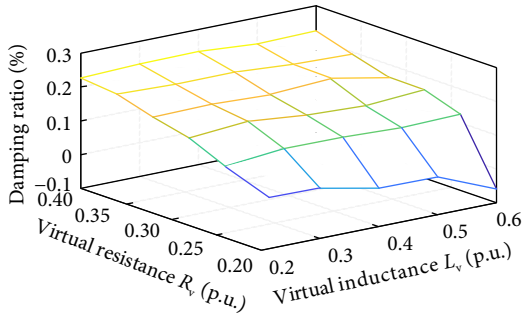


Fig. 21 Effects of changing virtual resistance and virtual inductance on the damping ratio of Mode 2, when the bandwidth of VBS loop is 50 Hz.

parameters. For a good compromise in the selection of controller parameters, two design parameters that are based on the damping ratios of dominant modes, i.e., the maximum ($\xi_1+\xi_2$) and the minimum ($\xi_1-\xi_2$), are introduced in this study.

Figure 22 provides a flowchart of the co-design method, which is elaborated step by step in the following:

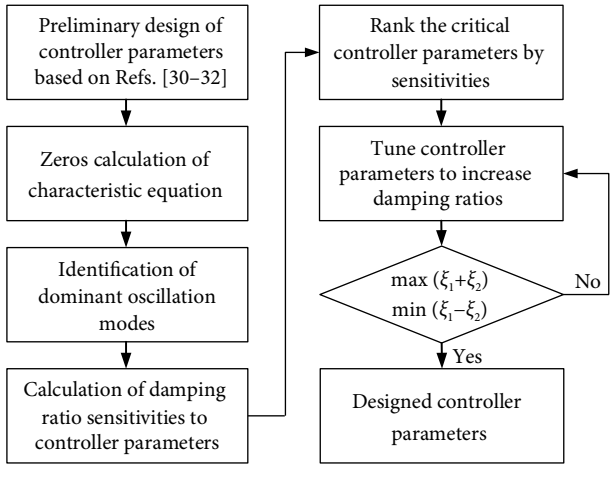


Fig. 22 Flowchart of the co-design method foe multiloop GFM control.

(1) Initial design of controller parameters using the analytical method from Refs. [27–29].

(2) Identification of dominant oscillation modes through the modal analysis (Section 3.2).

(3) Sensitivity analysis of damping ratios of dominant modes to controller parameters, resulting in critical control loops (Section 3.3).

(4) Analysis of the impact of critical control loops on dominant modes, i.e., Mode 1 and Mode 2 (Section 4.1).

(5) Assessment of the trade-off in tuning control parameters for high damping ratios at both Mode 1 and Mode 2.

(6) Tuning first the most critical controller parameter to yield the maximum ($\xi_1+\xi_2$) and the minimum ($\xi_1-\xi_2$)[30].

(7) Analysis of control-loop interactions by plotting damping ratios with changes of control parameters (Section 4.2).

(8) Final tuning of other controller parameters, based on the control-loop interaction analysis, for further higher ($\xi_1+\xi_2$) and lower ($\xi_1-\xi_2$).

5 Simulations and experimental validations

To verify the approach, nonlinear time-domain simulations and experiment tests are carried out, based on the parameters of the GFM inverter given in Table 1.

5.1 Simulations results

Figures 23–30 show the simulated waveforms of PoC voltage, active power, and reactive power with changes in different controller parameters. They illustrate how changes in these controller parameters affect the stability of the system and the damping ratios of the two oscillation modes.

Figure 23 shows that an increase in the integral gain of the AVC leads to system instability due to the decrease in the damping ratio of Mode 2. This correlates with the theoretical prediction in Figure 13. Figure 24 shows that an increase in the proportional gain of the PBS control from K_{ppsc} to $4K_{ppsc}$, also leads to instability. This aligns with the theoretical analysis in Figure 14.

Figures 25 and 26 depict the effects of changing the virtual resistance on the oscillation modes. A smaller virtual resistance ($0.5R_v$) manifests the oscillation Mode 2, while an increase in virtual resistance (to $1.5R_v$) triggers the oscillation Mode 1. These results agree with the theoretical analysis in Figure 17.

Figures 27 and 28 show the effects of changing the virtual inductance on the oscillation modes. An increase in virtual inductance from L_v to 1.5 L_v manifests the oscillation Mode 2, while a decrease in virtual inductance from L_v to 0.5 L_v induces the oscillation Mode 1. These results confirm the theoretical prediction in Figure 18.

Figure 29 shows that a decrease in the proportional gain of the current controller from K_{pc} to $0.5K_{pc}$ can also trigger the oscilaltion Mode 2. This is because a higher proportioanl gain provides a higher damping ratio at Mode 2.

Figure 30 shows that an increase in the proportional gain of VBS loop from K_{pv} to $1.5K_{pv}$. It is clear that the increase of K_{pv} also manifests the oscillation Mode 2, which agrees with the previous analysis in Figure 16.

5.2 Experimental results

The theoretical analysis and simulation results are further verified through experiments on a laboratory test setup. Figure 31 shows the layout of the experimental test setup. The Danfoss inverter is adopted as the GFM inverter. The DS1007 dSPACE system is

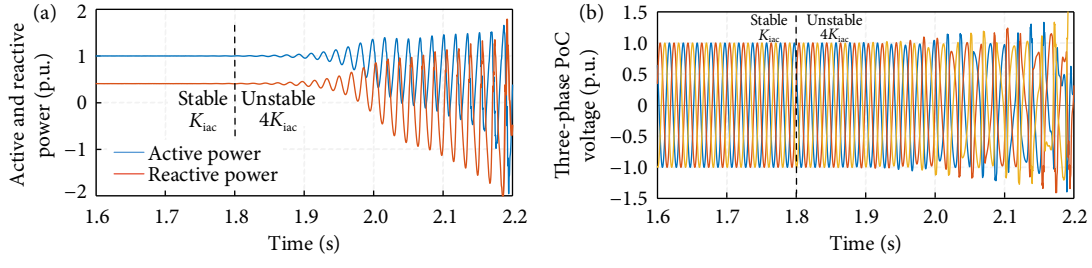


Fig. 23 Simulated waveforms of (a) PoC voltage, and (b) active power and reactive power, with different integral gains of AC voltage controller.

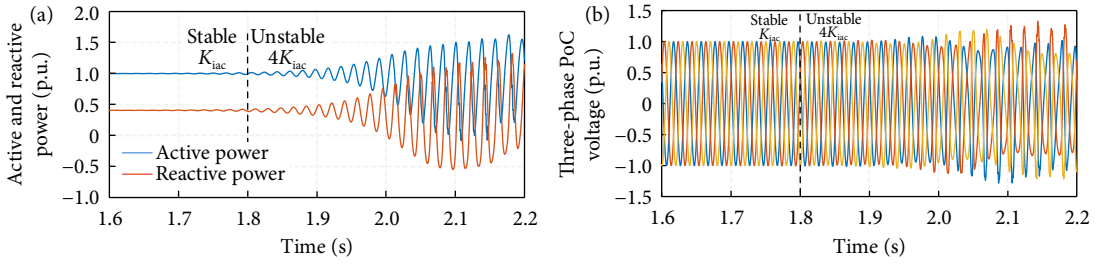


Fig. 24 Simulated waveforms of (a) PoC voltage, and (b) active power and reactive power, with different proportional gains of PBS control.

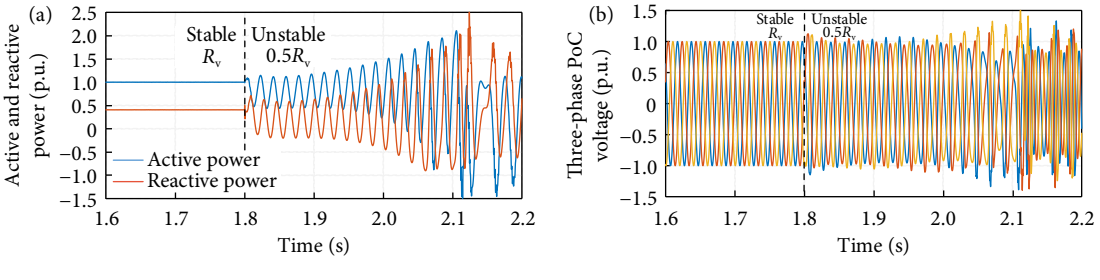


Fig. 25 Simulated waveforms of (a) PoC voltage, and (b) active power and reactive power, with the decrease of the virtual resistance.

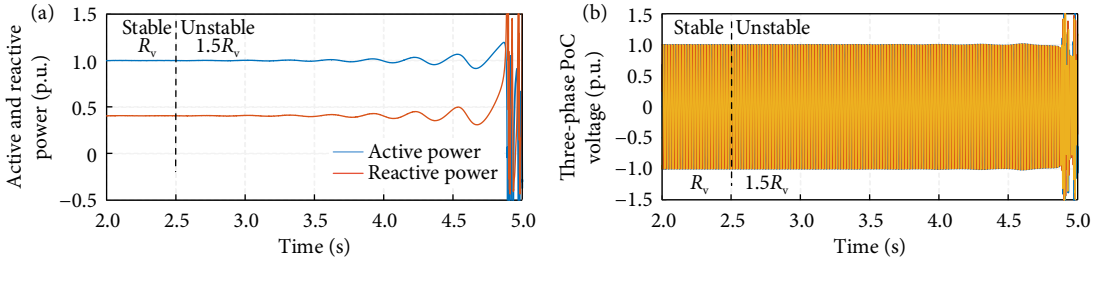


Fig. 26 Simulated waveforms of (a) PoC voltage, and (b) active power and reactive power, with the increase of virtual resistance.

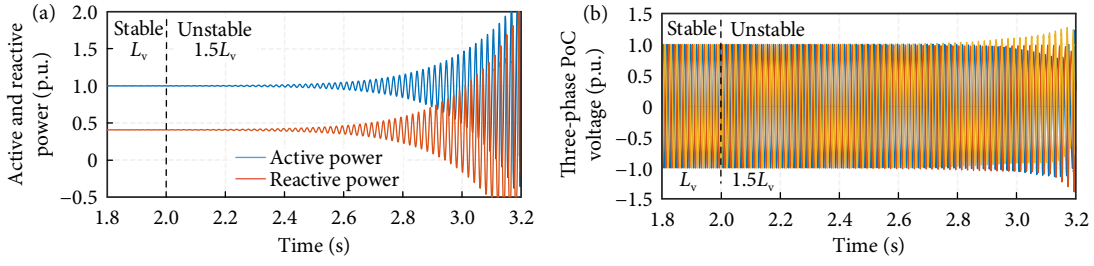


Fig. 27 Simulated waveforms of (a) PoC voltage, and (b) active power and reactive power, with the increase of virtual inductance.

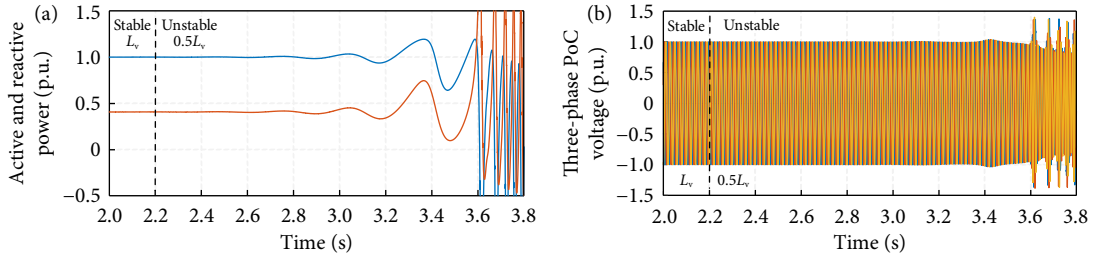


Fig. 28 Simulated waveforms of (a) PoC voltage, and (b) active power and reactive power, with the decrease of virtual inductance.

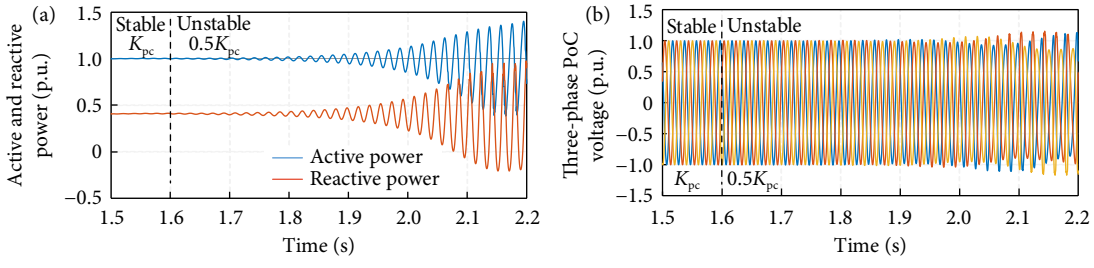


Fig. 29 Simulated waveforms of (a) PoC voltage, and (b) active power and reactive power, with the decrease of proportional gain of the current controller.

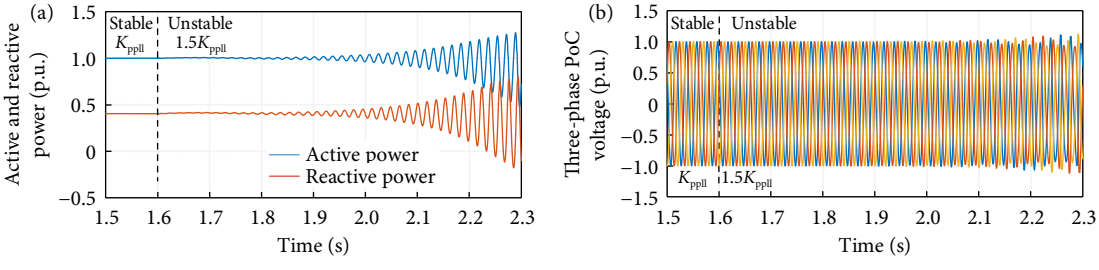


Fig. 30 Simulated waveforms of (a) PoC voltage, and (b) active power and reactive power, with the increase of proportional gain of VBS control.

used for implementing the control algorithms, where the DS5101 digital waveform output board is used to generate the gate driving pules for the inverter, and the DS2004 A/D board is used to measure the PoC voltage and inverter current. A programmable three-phase voltage source is used to emulate the power grid. The dc-link voltage of GFM inverter is kept at 730 V by an active rectifier.

All the system parameters used in experimetns are the same as that used in simulations. The line-to-line voltage at the PoC the per-phase current, the active power and reactive power of the inverter are measured in experiments.

The measured waveforms with different gains of AVC (Figure 32(a)) and PBS (Figure 32(b)) show that an increase in the gains cause a decrease in the damping ratio of Mode 2, and the system turns unstable. These resutls are consistent with simulation results in Figures 23 and 24, and the theoretical predictions in Figures 13 and 15.

The effects of changing the virtual resistance and the virtual inductance on the system are also tested, and the reuslts are shown in Figures 33 and 34. It is clear that a decrease in virtual resistance triggers the oscillation Mode 2, as shown in Figure 33(a),

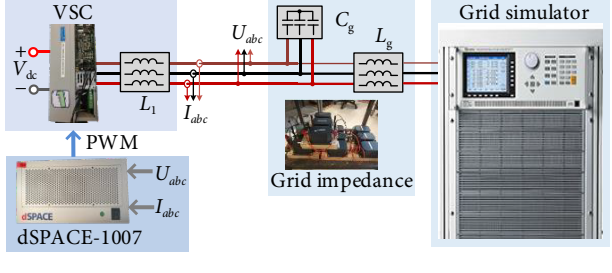


Fig. 31 Hardware picture of experiment setup.

while an increase in virtual resistance manifests the osciliation Mode 1, as shown in Figure 33(b). Another trade-off related to the virtual inductance is also observed in the test results shown in Figures 34(a) and 34(b). However, unlike the virtual resitance, the increase in virtual inductance decreases the damping ratio of Mode 2, while the decrease in virtual inductance decreases the damping ratio of Mode 1. These observations are aligned with the previous simulations results shown in Figures 27 and 28.

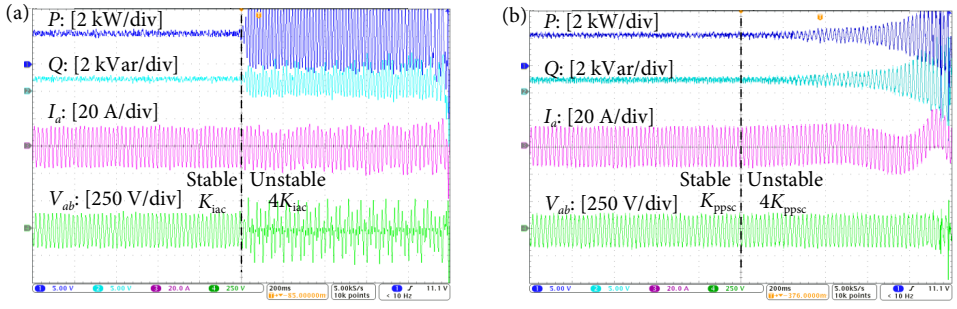


Fig. 32 Measured waveforms of PoC voltage, current, active power, and reactive power (a) with the increase of integral gain of AC voltage controller, (b) with the increase of proportional gain of PBS control.

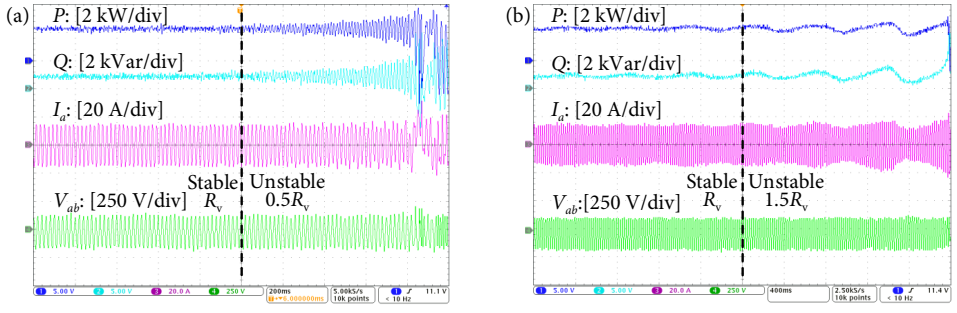


Fig. 33 Measured waveforms of PoC voltage, current, active power, and reactive power (a) with the decrease of the virtual resistance, (b) with the increase of virtual resistance.

Figure 35(a) validates that a decrease in proportional gain of the current controller manifests the oscillation Mode 2, which is consistent with the result in Figure 29.

Figure 35(b) shows the measured results with an increase in the proportional gain (from K_{pv} to $1.5K_{pv}$) of the VBS control. They are aligned with the theoretical prediction and simulation result shown in Figure 30.

6 Conclusions

This paper has discussed a design-oriented analysis approach for GFM inverters with the multi-loop control scheme. The control-loop interactions have been characterized through the frequency-domain modal analysis and parametric sensitivity analysis, showing how the tuning of one controller parameter affects the stability

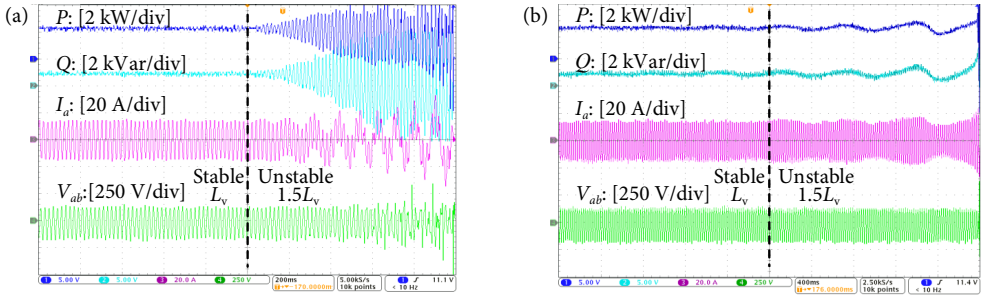


Fig. 34 Measured waveforms of PoC voltage, current, active power, and reactive power with (a) the increase of virtual inductance, and (b) the decrease of the virtual inductance.

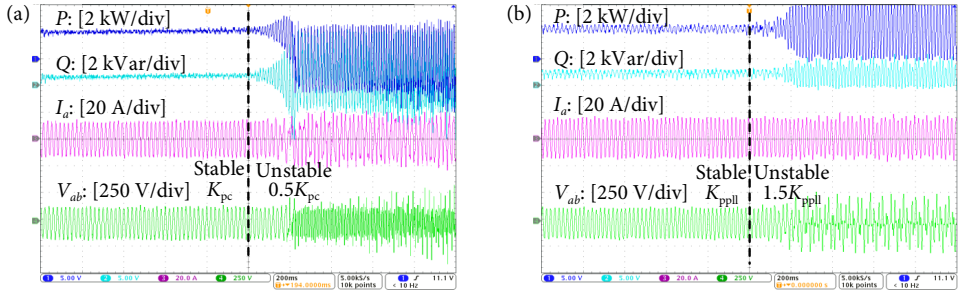


Fig. 35 Measured waveforms of PoC voltage, current, active power, and reactive power with (a) the decrease of proportional gain of the current controller, and (b) the increase of proportional gain of VBS control.

effect of others. A co-design method that considers the interactive effects is then discussed for the multiloop GFM control. The issue that tuning one controller parameter may have the opposite effect on the dominant modes has also been addressed in the design method. Both simulations and experimental results have validated the effectiveness of the design-oriented analysis method.

Appendix

A. Current control and virtual admittance model

For simplicity, the impedance modelling of the GFM inverter is performed step by step, beginning with the small-signal model of the current-vector control loop. Figure A1 depicts the block diagram of the current loop only, where the admittance model of the *L*-filter plant in the *dq*-frame is expressed as

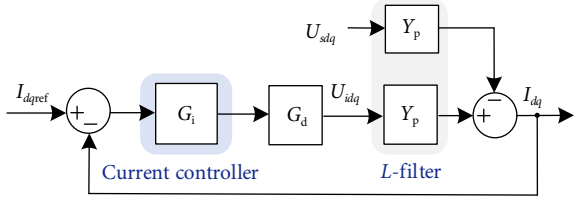


Fig. A1 Block diagram of the current controller.

$$\boldsymbol{Y}_p = \begin{bmatrix} sL_f & -\omega_1 L_f \\ \omega_1 L_f & sL_f \end{bmatrix}^{-1} \tag{A1}$$

By applying KVL across the filter admittance, the inverter output current in the *dq*-frame can be derived as

$$\boldsymbol{I}_{dq} = \boldsymbol{Y}_p(\boldsymbol{U}_{idq} - \boldsymbol{U}_{sdq}) \tag{A2}$$

where $\boldsymbol{U}_{idq}=[U_{id}, U_{iq}]^T$ represents the inverter output voltage in the *dq*-frame.

The current controller output is used as the reference voltage for the PWM generation. The inverter output voltage can be represented as

$$\boldsymbol{U}_{idq} = \boldsymbol{G}_d\boldsymbol{G}_i(\boldsymbol{I}_{dqref} - \boldsymbol{I}_{dq}) \tag{A3}$$

where $\boldsymbol{G}_i$ is the PI controller, which is given by

$$\boldsymbol{G}_i = \begin{bmatrix} K_{pc} + \dfrac{K_{ic}}{s} & 0 \\ 0 & K_{pc} + \dfrac{K_{ic}}{s} \end{bmatrix} \tag{A4}$$

where K_{pc} and K_{ic} mean the proportional gain and integral gain of the PI controller, respectively.

$\boldsymbol{G}_d$ is the time delay of the digital control system, which consists of one sampling period computation delay (T_s) and a half-sampling period ($0.5T_s$) of pulse width modulation delay[22]

$$\boldsymbol{G}_d = \begin{bmatrix} e^{-1.5T_s s} & 0 \\ 0 & e^{-1.5T_s s} \end{bmatrix} \tag{A5}$$

The closed-loop response of the current control loop can be derived based on Eqs. (A1)–(A5), which is given as

$$\boldsymbol{I}_{dq} = [\boldsymbol{I} + \boldsymbol{Y}_p\boldsymbol{G}_d\boldsymbol{G}_i]^{-1}\boldsymbol{Y}_p\boldsymbol{G}_d\boldsymbol{G}_i\boldsymbol{I}_{dqref} - [\boldsymbol{I} + \boldsymbol{Y}_p\boldsymbol{G}_d\boldsymbol{G}_i]^{-1}\boldsymbol{Y}_p\boldsymbol{U}_{sdq} \tag{A6}$$

The current references of GFM control $\boldsymbol{I}_{dqref}$ are generated through the virtual admittance and PoC voltage, which can be written as

$$\boldsymbol{I}_{dqref} = \boldsymbol{Y}_v(\boldsymbol{E}_{dq} - \boldsymbol{U}_{sdq}) = \underbrace{\begin{bmatrix} sL_v + R_v & -\omega_1 L_v \\ \omega_1 L_v & sL_v + R_v \end{bmatrix}^{-1}}_{\boldsymbol{Y}_v} \underbrace{\begin{bmatrix} E_d - U_{sd} \\ E_q - U_{sq} \end{bmatrix}}_{\boldsymbol{E}_{dq} - \boldsymbol{U}_{sdq}} \tag{A7}$$

where R_v and L_v represent the resistance and inductance of the virtual admittance, respectively. $\boldsymbol{E}_{dq}$ means the virtual electromotive force (EMF) in the *dq*-frame.

The EMF can be naturally expressed in the *dq*-frame as

$$\begin{cases} E_d = |E|(\cos^2\theta + \sin^2\theta) = |E| \\ E_q = |E|(\cos\theta\sin\theta - \sin\theta\cos\theta) = 0 \end{cases} \tag{A8}$$

where $|E|$ is the *d*-axis component of EMF, which is obtained from the output of AVC.

B. HSC

The Park (*dq*-) transformation has been used to transform three-phase variables into *dq*-frame, which may introduce synchronization dynamics on the transformed voltages and currents and influence the small-signal model of the GFM inverter. Since the controller of GFM inverter is implemented in the *dq*-frame that is aligned with EMF $\boldsymbol{E}$ and the GFM inverter is modelled in the *dq*-frame that is aligned with PoC voltage $\boldsymbol{U}_{sabc}$, the GFM inverter has two *dq*-frames: one is the system *dq*-frame defined by the PoC voltage, and another is the controller *dq*-frame defined by EMF, as shown in Figure B1. The phase angle difference between the controller *dq*-frame and the system *dq*-frame is $\theta_0+\Delta\theta$, where θ_0 denotes the phase angle difference between θ_{PoC} and θ in the steady-state. The phase difference $\Delta\theta$ is caused by the dynamics of the hybrid synchronization controller.

The voltage and current vectors used for the feedback control are transformed from the system *dq*-frame to the controller *dq*-frame through the matrix of $\boldsymbol{T}_d$. The modulated voltage generated by the feedback control are then transformed back to the system *dq*-frame with the inverse of matrix $\boldsymbol{T}_d$ to control the power semiconductors of the GFM inverter. The matrix of $\boldsymbol{T}_d$ can be expressed as

$$\boldsymbol{T}_d = \begin{bmatrix} \cos(\theta_{PoC} - \theta) & \sin(\theta_{PoC} - \theta) \\ -\sin(\theta_{PoC} - \theta) & \cos(\theta_{PoC} - \theta) \end{bmatrix} \tag{B1}$$

$$\boldsymbol{U}^s_{sdq} = \boldsymbol{T}_d\boldsymbol{U}^s_{sdq}; \boldsymbol{I}^s_{dq} = \boldsymbol{T}_d\boldsymbol{I}^s_{dq}; \boldsymbol{U}^s_{cdq} = \boldsymbol{T}_d\boldsymbol{U}^s_{cdq} \tag{B2}$$

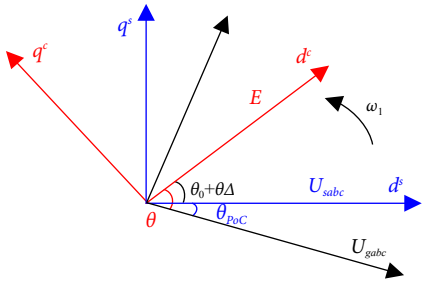


Fig. B1 System and controller *dq*-frames.

where the superscript "*c*" and "*s*" mean the variables in the controller *dq*-frame and system *dq*-frame, respectively.

Based on Figure 2, the synchronization phase angle θ obtained from the hybrid synchronization controller can be expressed as

$$\theta = \frac{1}{s}[\omega_1 + K_{pv}U_s\sin(\theta_{PoC} - \theta) + G_{PSC}(P_{ref} - P)] \tag{B3}$$

where K_{pv}means the proportional gain of VBS control; P_{ref} and P represent the reference and measurement of the active power, respectively; G_{PSC} represents the transfer function of PBS, where the PI controller is adopted to control the active power, which can be expressed as

$$G_{\mathrm{PSC}} = K_{\mathrm{ppsc}} + \frac{K_{\mathrm{ipsc}}}{s} \tag{B4}$$

where K_{ppsc} and K_{ipsc} denote the proportional and integral gain of the PI controller for PSC, respectively.

By adding a small-signal perturbation to Eqs. (B1) and (B3) and doing a small angle approximation of trigonometric functions, and cancelling the steady-state values, the small-signal model of synchronization phase angle can be written as

$$\Delta\theta = \frac{-G_{\mathrm{PSC}}\Delta P + K_{\mathrm{pv}}\left(-\sin\theta_0 U^s_{sd} + \cos\theta_0 U^s_{sq}\right)}{s + K_{\mathrm{pv}} U^s_{sd0}} \tag{B5}$$

where the active power difference ΔP is obtained based on the active power balance theory, which can be expressed as

$$\Delta P = U^s_{sd0} I^s_{\mathrm{d}} + U^s_{sq0} I^s_{\mathrm{q}} + I^s_{sd0} U^s_{\mathrm{d}} + I^s_{sq0} U^s_{\mathrm{q}} \tag{B6}$$

where the subscript "0" represents the steady-state value of the corresponding variable.

Based on Figure 40, the relationships between vectors in the controller dq-frame and the vectors in the system dq-frame can be expressed as

$$U^c_{sdq} = \begin{bmatrix} \cos(\theta_0+\Delta\theta) & \sin(\theta_0+\Delta\theta) \\ -\sin(\theta_0+\Delta\theta) & \cos(\theta_0+\Delta\theta) \end{bmatrix} U^s_{sdq} \tag{B7}$$

$$I^c_{dq} = \begin{bmatrix} \cos(\theta_0+\Delta\theta) & \sin(\theta_0+\Delta\theta) \\ -\sin(\theta_0+\Delta\theta) & \cos(\theta_0+\Delta\theta) \end{bmatrix} I^s_{dq} \tag{B8}$$

$$U^s_{cdq} = \begin{bmatrix} \cos(\theta_0+\Delta\theta) & -\sin(\theta_0+\Delta\theta) \\ \sin(\theta_0+\Delta\theta) & \cos(\theta_0+\Delta\theta) \end{bmatrix} U^c_{cdq} \tag{B9}$$

By doing a small angle approximation of trigonometric functions, and cancelling the steady-state values, the relationship between the voltage vector in the control dq-frame and system dq-frame can be further written as

$$\begin{bmatrix} U^c_{sd} \\ U^c_{sq} \end{bmatrix} = \underbrace{\begin{bmatrix} \cos\theta_0 & \sin\theta_0 \\ -\sin\theta_0 & \cos\theta_0 \end{bmatrix}}_{T} \left(\begin{bmatrix} U^s_{sd} \\ U^s_{sq} \end{bmatrix} - \begin{bmatrix} -U^s_{sq0}\Delta\theta \\ U^s_{sd0}\Delta\theta \end{bmatrix} \right) \tag{B10}$$

By substituting the expression of phase angle (23) into (25), the expression can be derived as

$$\begin{aligned} \begin{bmatrix} U^c_{sd} \\ U^c_{sq} \end{bmatrix} &= \begin{bmatrix} \cos\theta_0 & \sin\theta_0 \\ -\sin\theta_0 & \cos\theta_0 \end{bmatrix} \left(\begin{bmatrix} U^s_{sd} \\ U^s_{sq} \end{bmatrix} - \begin{bmatrix} -U^s_{sq0}\Delta\theta \\ U^s_{sd0}\Delta\theta \end{bmatrix} \right) \\ &= \boldsymbol{T}\left(\begin{bmatrix} U^s_{sd} \\ U^s_{sq} \end{bmatrix} - \boldsymbol{H}_{11} \begin{bmatrix} I^s_{\mathrm{d}} \\ I^s_{\mathrm{q}} \end{bmatrix} - \boldsymbol{H}_{12} \begin{bmatrix} U^s_{sd} \\ U^s_{sq} \end{bmatrix} \right) \end{aligned} \tag{B11}$$

where

$$\boldsymbol{H}_{11} = \begin{bmatrix} \dfrac{G_{\mathrm{PLC}} U^s_{sd0} U^s_{sq0}}{s + K_{\mathrm{pv}} U^s_{sd0}} & \dfrac{G_{\mathrm{PLC}} U^s_{sq0} U^s_{sq0}}{s + K_{\mathrm{pv}} U^s_{sd0}} \\ -\dfrac{G_{\mathrm{PLC}} U^s_{sd0} U^s_{sd0}}{s + K_{\mathrm{pv}} U^s_{sd0}} & -\dfrac{G_{\mathrm{PLC}} U^s_{sd0} U^s_{sq0}}{s + K_{\mathrm{pv}} U^s_{sd0}} \end{bmatrix},$$

$$\boldsymbol{H}_{12} = \begin{bmatrix} \dfrac{G_{\mathrm{PLC}} I^s_{sd0} U^s_{sq0} + U^s_{sq0} K_{\mathrm{pv}} \sin\theta_0}{s + K_{pv} U^s_{sd0}} \\ \dfrac{G_{\mathrm{PLC}} U^s_{sq0} I^s_{sq0} - U^s_{sq0} K_{\mathrm{pv}} \cos\theta_0}{s + K_{pv} U^s_{sd0}} \\ \dfrac{-G_{\mathrm{PLC}} I^s_{sd0} U^s_{sd0} - U^s_{sd0} K_{pv} \sin\theta_0}{s + K_{\mathrm{pv}} U^s_{sd0}} \\ \dfrac{-G_{\mathrm{PLC}} U^s_{sd0} I^s_{sq0} + U^s_{sd0} K_{pv} \cos\theta_0}{s + K_{\mathrm{pv}} U^s_{sd0}} \end{bmatrix}$$

Similarly, the relationship between the current in the control dq-frame and system dq-frame can be expressed by

$$\begin{aligned} \begin{bmatrix} I^c_d \\ I^c_q \end{bmatrix} &= \underbrace{\begin{bmatrix} \cos\theta_0 & \sin\theta_0 \\ -\sin\theta_0 & \cos\theta_0 \end{bmatrix}}_{T} \left(\begin{bmatrix} I^s_d \\ I^s_q \end{bmatrix} - \begin{bmatrix} -I^s_{q0}\Delta\theta \\ I^s_{d0}\Delta\theta \end{bmatrix} \right) \\ &= \boldsymbol{T}\left(\begin{bmatrix} I^s_d \\ I^s_q \end{bmatrix} - \boldsymbol{H}_{21} \begin{bmatrix} I^s_d \\ I^s_q \end{bmatrix} - \boldsymbol{H}_{22} \begin{bmatrix} U^s_{sd} \\ U^s_{sq} \end{bmatrix} \right) \end{aligned} \tag{B12}$$

where

$$\boldsymbol{H}_{21} = \begin{bmatrix} \dfrac{G_{\mathrm{PLC}} U^s_{sd0} I^s_{sq0}}{s + K_{\mathrm{pv}} U^s_{sd0}} & \dfrac{G_{\mathrm{PLC}} U^s_{sq0} I^s_{sq0}}{s + K_{\mathrm{pv}} U^s_{sd0}} \\ -\dfrac{G_{\mathrm{PLC}} U^s_{sd0} I^s_{sd0}}{s + K_{\mathrm{pv}} U^s_{sd0}} & -\dfrac{G_{\mathrm{PLC}} I^s_{sd0} U^s_{sq0}}{s + K_{\mathrm{pv}} U^s_{sd0}} \end{bmatrix},$$

$$\boldsymbol{H}_{22} = \begin{bmatrix} \dfrac{G_{\mathrm{PLC}} I^s_{sd0} I^s_{sq0} + I^s_{sq0} K_{\mathrm{pv}} \sin\theta_0}{s + K_{\mathrm{pv}} U^s_{sd0}} \\ \dfrac{G_{\mathrm{PLC}} I^s_{sq0} I^s_{sq0} - I^s_{sq0} K_{\mathrm{pv}} \cos\theta_0}{s + K_{\mathrm{pv}} U^s_{sd0}} \\ \dfrac{-G_{\mathrm{PLC}} I^s_{sd0} I^s_{sd0} - I^s_{sd0} K_{\mathrm{pv}} \sin\theta_0}{s + K_{\mathrm{pv}} U^s_{sd0}} \\ \dfrac{-G_{\mathrm{PLC}} I^s_{sd0} I^s_{sq0} + U^s_{sd0} K_{\mathrm{pv}} \cos\theta_0}{s + K_{\mathrm{pv}} U^s_{sd0}} \end{bmatrix} \tag{B13}$$

Through the Park inverse transformation, the relationship between the modulated voltage in the control dq-frame and system dq-frame can be written as

$$\begin{aligned} \begin{bmatrix} U^s_{cd} \\ U^s_{cq} \end{bmatrix} &= \underbrace{\begin{bmatrix} \cos\theta_0 & -\sin\theta_0 \\ \sin\theta_0 & \cos\theta_0 \end{bmatrix}}_{T^{-1}} \left(\begin{bmatrix} U^c_{cd} \\ U^c_{cq} \end{bmatrix} + \begin{bmatrix} -U^s_{cq0}\Delta\theta \\ U^c_{cd0}\Delta\theta \end{bmatrix} \right) = \\ &\boldsymbol{T}^{-1}\left(\begin{bmatrix} U^c_{cd} \\ U^c_{cq} \end{bmatrix} + \boldsymbol{H}_{31} \begin{bmatrix} I^s_d \\ I^s_q \end{bmatrix} + \boldsymbol{H}_{32} \begin{bmatrix} U^s_{sd} \\ U^s_{sq} \end{bmatrix} \right) \end{aligned} \tag{B14}$$

where

$$\boldsymbol{H}_{31} = \begin{bmatrix} \dfrac{G_{\mathrm{PLC}} U^s_{sd0} U^c_{cq0}}{s + K_{\mathrm{pv}} U^s_{sd0}} & \dfrac{G_{\mathrm{PLC}} U^s_{sq0} U^c_{cq0}}{s + K_{\mathrm{pv}} U^s_{sd0}} \\ -\dfrac{G_{\mathrm{PLC}} U^s_{sd0} U^c_{cd0}}{s + K_{\mathrm{pv}} U^s_{sd0}} & -\dfrac{G_{\mathrm{PLC}} U^c_{cd0} U^s_{sq0}}{s + K_{\mathrm{pv}} U^s_{sd0}} \end{bmatrix},$$

$$\boldsymbol{H}_{32} = \begin{bmatrix} \dfrac{G_{\mathrm{PLC}} I^s_{sd0} U^c_{cq0} + U^c_{cq0} K_{pv} \sin\theta_0}{s + K_{\mathrm{pv}} U^s_{sd0}} \\ \dfrac{G_{\mathrm{PLC}} I^s_{sq0} U^c_{cq0} - U^c_{cq0} K_{\mathrm{pv}} \cos\theta_0}{s + K_{\mathrm{ppll}} U^s_{sd0}} \\ \dfrac{-G_{\mathrm{PLC}} I^s_{sd0} U^c_{cd0} - U^c_{cd0} K_{\mathrm{pv}} \sin\theta_0}{s + K_{\mathrm{pv}} U^s_{sd0}} \\ \dfrac{-G_{\mathrm{PLC}} U^c_{cd0} I^s_{sq0} + U^s_{sd0} K_{\mathrm{pv}} \cos\theta_0}{s + K_{\mathrm{pv}} U^s_{sd0}} \end{bmatrix}$$

C. AVC Model

The magnitude of the PoC voltage is controlled to provide the d-axis component of the EMF, which can be written as

$$|E| = \frac{K_{\mathrm{iac}}}{s}\left(U_{\mathrm{ref}} - \sqrt{{U^s_{sd}}^2 + {U^s_{sq}}^2}\right) \tag{C1}$$

where U_{ref} represents the reference of PoC voltage magnitude; K_{iac} denotes the integral gain of the AC voltage magnitude controller. By adding a small-signal perturbation to (C1), the small-signal model can be obtained as

$$|E| = \underbrace{\begin{bmatrix} \dfrac{-K_{\mathrm{iac}} U^s_{sd0}}{s|U_s|} & \dfrac{-K_{iac} U^s_{sd0}}{s|U_s|} \\ 0 & 0 \end{bmatrix}}_{G_a} \boldsymbol{U}^s_{sdq} \tag{C2}$$

where $\boldsymbol{G}_a$ is the transfer function matrix of the AC voltage magnitude controller.

Article history

Received: 26 October 2022; Revised: 5 February 2023; Accepted: 17 February 2023

Additional information

© 2023 The Author(s). This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

Declaration of competing interest

The authors have no competing interests to declare that are relevant to the content of this article.

References

[1] Lasseter, R. H., Chen, Z., Pattabiraman, D. (2020). Grid-forming inverters: A critical asset for the power grid. *IEEE Journal of Emerging and Selected Topics in Power Electronics*, 8: 925–935.

[2] Matevosyan, J., Vital, V., O'Sullivan, J., Quint, R., Badrzadeh, B., Prevost, T., Quitmann, E., Ramasubramanian, D., Urdal, H., Achilles, S., MacDowell, J., Huang, S. H. (2019). Grid-forming inverters: Are they the key for high renewable penetration. *IEEE Power and Energy Magazine*, 17: 89–98.

[3] Wang, X., Taul, M. G., Wu, H., Liao, Y., Blaabjerg, F., Harnefors, L. (2020). Grid-synchronization stability of Converter-based resources—an overview. *IEEE Open Journal of Industry Applications*, 1: 115–133.

[4] Zhang, L., Harnefors, L., Nee, H. (2010). Power-synchronization control of grid-connected voltage-source converters. *IEEE Transactions on Power Systems*, 25: 809–820.

[5] Denis, G., Prevost, T., Panciatici, P., Kestelyn, X., Colas, F., Guillaud, X. (2016). Improving robustness against grid stiffness, with internal control of an AC voltage-controlled VSC. In: Proceedings of the 2016 IEEE Power and Energy Society General Meeting (PESGM), Boston, MA, USA.

[6] Harnefors, L., Kukkola, J., Routimo, M., Hinkkanen, M., Wang, X. (2021). A universal controller for grid-connected voltage-source converters. *IEEE Journal of Emerging and Selected Topics in Power Electronics*, 9: 5761–5770.

[7] Huang, L., Xin, H., Wang, Z., Zhang, L., Wu, K., Hu, J. (2019). Transient stability analysis and control design of droop-controlled voltage source converters considering current limitation. *IEEE Transactions on Smart Grid*, 10: 578–591.

[8] Harnefors, L., Schweizer, M., Kukkola, J., Routimo, M., Hinkkanen, M., Wang, X. (2022). Generic PLL-based grid-forming control. *IEEE Transactions on Power Electronics*, 37: 1201-1204.

[9] Rodriguez, P., Candela, I., Citro, C., Rocabert, J., Luna, A. (2013). Control of grid-connected power converters based on a virtual admittance control loop. In: Proceedings of the 2013 15th European Conference on Power Electronics and Applications (EPE), Lille, France.

[10] Rodríguez, P., Citro, C., Candela, J. I., Rocabert, J., Luna, A. (2018). Flexible grid connection and islanding of SPC-based PV power converters. *IEEE Transactions on Industry Applications*, 54: 2690–2702.

[11] Taul, M. G., Wang, X., Davari, P., Blaabjerg, F. (2020). Current limiting control with enhanced dynamics of grid-forming converters during fault conditions. *IEEE Journal of Emerging and Selected Topics in Power Electronics*, 8: 1062–1073.

[12] Barklund, E., Pogaku, N., Prodanovic, M., Hernandez-Aramburo, C., Green, T. C. (2008). Energy management in autonomous microgrid using stability-constrained droop control of inverters. *IEEE Transactions on Power Electronics*, 23: 2346–2352.

[13] Liao, Y., Wang X., Blaabjerg, F. (2020). Passivity-based analysis and design of linear voltage controllers for voltage-source Converters. *IEEE Open Journal of the Industrial Electronics Society*, 1: 114–126.

[14] Yu, K., Ai, Q., Wang, S., Ni, J., Lv, T. (2016). Analysis and optimization of droop controller for microgrid system based on small-signal dynamic model. *IEEE Transactions on Smart Grid*, 7: 695–705.

[15] Rosso, R., Engelken, S., Zou, Z., Willich, V., Liserre, M. (2018). Parameter sensitivity analysis of SPC-based control under different grid conditions. In: Proceedings of the IECON 2018 - 44th Annual Conference of the IEEE Industrial Electronics Society, Washington, DC, USA.

[16] Marin, L., Tarras´, A., Candela, I., Rodriguez, P. (2018). Stability analysis of a grid-connected VSC controlled by SPC. In: Proceedings of the International Conference on Renewable Energy Research and Applications (ICRERA), Paris, France.

[17] Liao, Y., Wang, X., Liu, F., Xin, K., Liu, Y. (2020). Sub-synchronous control interaction in grid-forming VSCs with droop control. In: Proceedings of the 2019 4th IEEE Workshop on the Electronic Grid (eGRID), Xiamen, China.

[18] Dokus, M., Mertens, A. (2021). On the coupling of power-related and inner inverter control loops of grid-forming converter systems. *IEEE Access*, 9: 16173–16192.

[19] Qoria, T., Gruson, F., Colas, F., Guillaud, X., Debry, M. S., Prevost, T. (2018). Tuning of cascaded controllers for robust grid-forming voltage source converter. In: Proceedings of the 2018 Power Systems Computation Conference (PSCC), Dublin, Ireland.

[20] Gong, H., Wang, X., Harnefors, L. (2021). Rethinking Current controller design for PLL-synchronized VSCs in weak grids. *IEEE Transactions on Power Electronics*, 37: 1369–1381.

[21] Wang, X., Harnefors, L., Blaabjerg, F. (2018). Unified impedance model of grid-connected voltage-source converters. *IEEE Transactions on Power Electronics*, 33: 1775–1787.

[22] Liu S., Dougal, R. A. (2002). Dynamic multiphysics model for solar array. *IEEE Transactions on Energy Conversion*, 17: 285–294.

[23] Xu, L., Yao, L., Sasse, C. (2007). Grid integration of large DFIG-based wind farms using VSC transmission. *IEEE Transactions on Power Systems*, 22: 976–984.

[24] Zhang, H., Harnefors, L., Wang, X., Gong, H., Hasler, J. (2019). Stability analysis of grid-connected voltage-source converters using SISO modelling. *IEEE Transactions on Power Electronics*, 34: 8104–8117.

[25] Zhan, Y., Xie, X., Liu, H., Liu, H., Li, Y. (2019). Frequency-domain modal analysis of the oscillatory stability of power systems with high-penetration renewables. *IEEE Transactions on Sustainable Energy*, 10: 1534–1543.

[26] Zhao, F., Wang, X., Zhou, Z., Harnefors, L., Svensson, J. R., Kocewiak, Ł. H., Peter Sidoroff Gryning, M. (2022). Control interaction modeling and analysis of grid-forming battery energy storage system for offshore wind power plant. *IEEE Transactions on Power Systems*, 37: 497–507.

[27] Wu, H., Ruan, X., Yang, D., Chen, X., Zhao, W., Lv, Z., Zhong, Q.C. (2016). Small-signal modeling and parameters design for virtual synchronous generators. *IEEE Transactions on Industrial Electronics*, 63: 4292–4303.

[28] Harnefors, L., Hinkkanen, M., Riaz, U., Mahafugur Rahman, F. M., Zhang, L. (2019). Robust analytic design of power-synchronization control. *IEEE Transactions on Industrial Electronics*, 66: 5810–5819.

[29] Zhang, H., Harnefors, L., Wang, X., Hasler, J. P., Ostlund, S., Danielsson, C., Gong, H. (2021). Loop-at-a-time stability analysis for grid-connected voltage-source converters. *IEEE Journal of Emerging and Selected Topics in Power Electronics*, 9: 5807–5821.

[30] D'Arco, S., Suul, J. A., Fosso, O. B. (2014). Automatic tuning of cascaded controllers for power converters using eigenvalue parametric sensitivities. *IEEE Transactions on Industry Applications*, 51: 1743–1753.